Impact Absorbing and Compensation for Heavy Object Catching With an Unmanned Aerial Manipulator

Siqiang Wang, Zongyu Ma, Fengyu Quan, and Haoyao Chen, *Member, IEEE*

***Abstract*—Maintaining stability after catching an object is essential for the flight safety of unmanned aerial manipulators (UAMs). Current research on UAM catching stationary or moving targets neglects the impact of the target and the introduced center of mass (COM) displacement. This letter proposes an approach to absorb the impact during the capture of fast-flying heavy targets by UAMs, ultimately reducing UAM movement. An angular momentum theorem-based fast target mass estimation algorithm is proposed to estimate the target mass within the limited catching time. With the estimated target mass, a mass compensated impedance controller for a brushless motor-based arm and a compensation algorithm for the UAM attitude controller are proposed. An innovative aspect of our approach is utilizing the estimated target mass to adjust the impedance controller and compensate for COM displacement, which enables effective absorption of the target's impact and reduces UAM movement. We validate our proposed approach through simulation and real-world experiments. The results demonstrate that our approach significantly reduces UAM body displacement after capturing fast-flying heavy targets, enhancing UAM's flight safety.**

***Index Terms*—Aerial systems: mechanics and control, compliance and impedance control, robot safety.**

I. Introduction

THE unmanned aerial manipulator, consisting of an unmanned aerial vehicle (UAV) and one or more robotic arms, is a suitable type of aerial robot for tasks that involve physical interactions with the environment, as supported by previous research [1], [2], [3]. Among various applications in different fields, catching and maneuvering a heavy and moving object is considered one of the most challenging tasks for an aerial manipulator. According to the analysis in [4], recent target-catching research has focused on the precise capture of static or quasi-static light targets [5], [6], in which the end effector and target are relatively static, and the target's momentum act on the UAM can be regarded as zero. The impact of the target and the introduced COM displacement should be addressed. This letter aims to provide an efficient approach for UAM to maintain stability while catching fast-flying heavy targets and maximize flight stability after the catching.

Bio-inspired applications require the UAM to catch a target while the UAM is moving, as shown in [7], [8], [9], [10], which introduced new challenges, including target state estimation, trajectory prediction, impact absorption, and target mass estimation. Catching a flying ball with a single UAV [5], [11], [12], [13] or multiple UAVs [14] has been achieved in the past. However, the above research did not consider the influence of the target's mass on the flight dynamics. Researchers modeled the impact process to catch a moving target on the ground with UAM and designed an estimator to estimate the target mass in [15]. Defining a quasi-static catching, the impact of the target is not significant enough. However, due to the dynamical limitation of the UAM and the task time limitation, keeping quasi-static between the UAM and the target is sometimes impossible. Hence, an impact-absorbing mechanism is needed. In addition to the immediate impact during catching, the target also influences the position and orientation tracking by introducing COM displacement [16].

The impedance controller is a common method to realize the impact-absorbing mechanism. Most UAM researchers employ high gear ratio servos as actuators for lightweight arms [17], [18], which suffer from backlashes [4] and high static friction due to the gear ratio. The drawbacks of servo make the implementation of torque controllers on servo-based UAM challenging, as accurate torque control becomes difficult.

Several methods are proposed to compensate for the influence of the COM displacement. One interesting method is a three-layer compensation consisting of COM displacement estimation, external force/torque estimation, and battery-movement-based compensation [16], [19]. The COM displacement the target introduces is considered an external force and estimated within seconds. Similar works such as [20], [21] also take several seconds to estimate the generalized external force or the target mass. However, given that the largest positional and angular displacements occur within 1 s after the catching time, the estimation process must be faster to ensure flight safety. A neural network adaptive force control algorithm is proposed

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The authors are with the School of Mechanical Engineering and Automation, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: hychen5@hit.edu.cn).

A video showcasing our experiments can be accessed at https://youtu.be/k28Xirbjuek.

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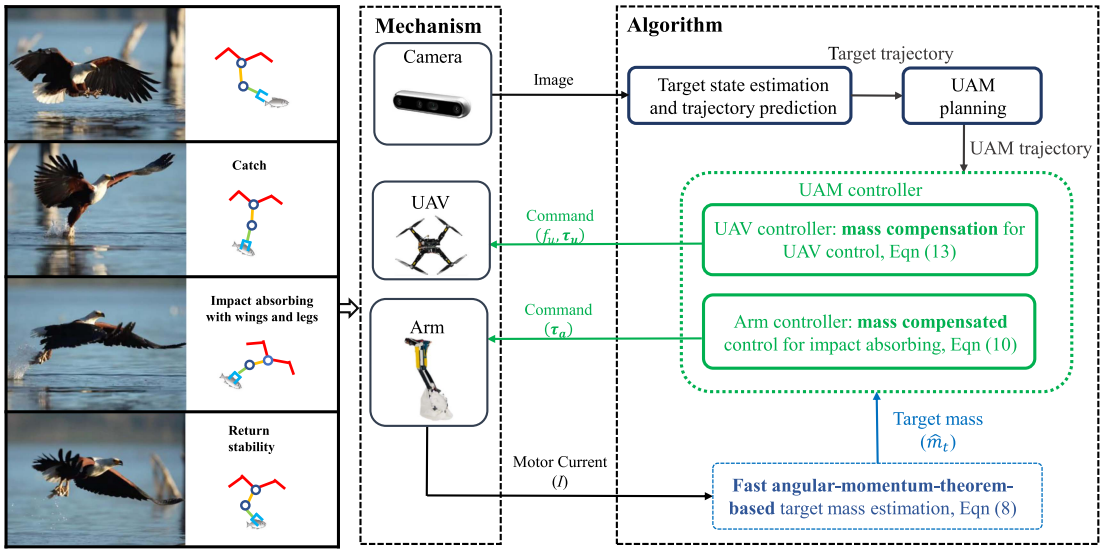


Fig. 1. Information flow of catching a heavy flying target with an aerial manipulator.

in [6], which can adapt to different mass targets. However, the neural network is trained offline, which is highly customized for the specified platform, and the generalization ability to different scenes and targets has yet to be tested in real-world applications.

Impact absorbing has also been studied in both fixed manipulator [22] and ground mobile manipulator [23]. Various passive and active methods have been designed to absorb the impact introduced by the target. The spring mechanism is the most used passive method [24], [25], [26], [27], known for its energy-saving and simplicity. However, the absorber's damping is fixed, incapable of dealing with different catching situations. Active impact-absorbing methods are usually implemented with the joint controller. For fixed robotic arms, the intuitive idea is to control the arm along the target trajectory after catching, as demonstrated in [28], [29], [30]. However, this requires predicting the target trajectory and ensuring the system can track it accurately. The idea is similar to quasi-static catching, which is only suitable for fast-flying targets in certain situations. For robotic arms mounted on unmanned ground vehicles, impedance controllers are widely used to catch flying targets [22], [31]. However, the impedance controller is not off the shelf for UAM since UAM's dynamics model is more complicated.

Summarizing the previous studies on catching fast and heavy targets with a UAM, most of them have yet to consider the post-catching stability of the UAM. It is still a big challenge for UAM to replicate the mimicking of a hawk catching a flying pigeon, in which the mass ratio between the target and the hawk is around 20–30%. Catching a fast and heavy target while maintaining stability with a UAM is still a challenging and interesting problem. Taking inspiration from the hunting behavior of hawks and the movements of humans catching fast and heavy objects, we develop an approach as depicted in Fig. 1. The approach helps absorb the impact introduced while catching fast-flying heavy targets and ensures flight safety by reducing movement and tracking errors. An impedance controller is employed for the UAM's arm to absorb the impact of the target. However, a fixed impedance setting cannot adapt to targets of different weights and velocities. Therefore, we propose a real-time algorithm to estimate the target's mass and compensate it in the impedance controller parameters accordingly. The estimated target mass is also used to compensate for the UAM pose controller, improving trajectory tracking performance.

To validate the proposed algorithms, we have developed an unmanned aerial manipulator platform implemented in simulated and real environments. The simulation environment is created using Gazebo, an ODE-based simulator widely used in robotics research. In the simulation, the main parameters of the UAM, such as UAV mass and arm length, are set to be consistent with those of the real UAM platform. A robot arm is designed and mounted on a UAV for the real platform, while a net is employed as the end effector for simplicity.

Our main contributions are fourfold.

- Firstly, we analyze and model the contact between the target and the UAM. Based on this contact model, we propose a novel fast target mass estimation algorithm capable of estimating the target mass within 0.1 seconds.
- Secondly, we design an impedance controller for the UAM arm in joint space. This controller effectively absorbs the target's kinetic energy and reduces the angular displacement of the UAM.
- Thirdly, we develop a compensation algorithm for the UAM's position controller. This algorithm mitigates the tracking error caused by the center of mass displacement after catching the target.
- Fourthly, by combining the components above, UAM successfully demonstrates the capability to catch a fast-flying target (moving at 10 m/s) that is considerably heavy (25% of the UAM's mass) meanwhile maintaining system stability.

The rest of this letter is structured as follows. After formulating the dynamic models of UAM and the problem in Section II, we present our approach designed for absorbing the impact and stabilizing the UAM in Section III. Section IV presents

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the experiments conducted in both simulation and real-world environments. Section V concludes the work of this letter and discusses future research directions.

II. Modeling and Problem Formulation

The timeline of UAM catching a flying target can be divided into three stages based on the time of catching t_0 and when UAM returns to stable state t_1. The first stage is under-catching, where $t < t_0$. The second stage is the contact stage, where $t_0 < t < t_1$ and $t_1 - t_0$ represent the impact absorbing time. The third stage is post-catching, where $t_1 < t$. The first and the third stages have been modeled and studied in previous works [5], [6]. This letter focuses on the contact stage $t_0 < t < t_1$.

A. UAM Dynamics

Let us define $\boldsymbol{\xi} = [\boldsymbol{x}_u^T, \boldsymbol{\Phi}_u^T, q_1, \ldots, q_n]^T \in \mathbb{R}^{n_\xi}$ as the generalized state of the UAM, where $\boldsymbol{x}_u = [p_x, p_y, p_z]^T$ are the position of the UAM, and $\boldsymbol{\Phi}_u = [\phi, \vartheta, \psi]^T$ are the Euler angles of the UAM, q_i is the joint angle of the i-th joint, and $n_\xi = 6 + n$ is the dimension of the generalized state.

As derived in Equations (10)–(18) of [32], we omit the Euler-Lagrange formulation for simplicity and directly introduce the dynamics of the UAM as

$$\boldsymbol{M}(\boldsymbol{\xi})\ddot{\boldsymbol{\xi}} + \boldsymbol{C}(\dot{\boldsymbol{\xi}}, \boldsymbol{\xi})\dot{\boldsymbol{\xi}} + \boldsymbol{G}(\boldsymbol{\xi}) = \boldsymbol{u} + \boldsymbol{u}_{ext}, \tag{1}$$

where $\boldsymbol{M}(\boldsymbol{\xi})$ is an $(n_\xi \times n_\xi)$ symmetric and positive definite inertia matrix, $\boldsymbol{C}(\dot{\boldsymbol{\xi}}, \boldsymbol{\xi})$ is the extended Coriolis matrix, $\boldsymbol{G}(\boldsymbol{\xi}) = \dfrac{\partial \mathcal{U}(\boldsymbol{\xi})}{\partial \boldsymbol{\xi}}$ is the gravity term, $\mathcal{U}$ is the potential energy of the system, $\boldsymbol{u}$ is the generalized control input, and $\boldsymbol{u}_{ext}$ is the external generalized force applied on the UAM. The elements of $\boldsymbol{M} = \begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{11} & \boldsymbol{M}_{13} \\ \boldsymbol{M}_{12}^T & \boldsymbol{M}_{22} & \boldsymbol{M}_{23} \\ \boldsymbol{M}_{13}^T & \boldsymbol{M}_{23}^T & \boldsymbol{M}_{33} \end{bmatrix}$ and $\boldsymbol{C} = \left[\boldsymbol{C}_1^T, \boldsymbol{C}_2^T, \boldsymbol{C}_3^T\right]^T$ can be found in [32], and $\boldsymbol{M}_{11} = (m_u + u_a)\boldsymbol{I}_3$. The generalized input $\boldsymbol{u}$ is defined as

$$\boldsymbol{u} = \bar{\boldsymbol{R}}_b \boldsymbol{f}, \tag{2}$$

where $\boldsymbol{f} = [0, 0, f_u, \boldsymbol{\tau}_u^T, \boldsymbol{\tau}_a^T]^T$, f_u and $\boldsymbol{\tau}_u = [\tau_\phi, \tau_\vartheta, \tau_\psi]^T \in \mathbb{R}^3$ are the total lift and the torque generated by the UAV motors, $\boldsymbol{\tau}_a \in \mathbb{R}^n$ is the arm torques generated by arm motors, $\bar{\boldsymbol{R}}_b = diag(\boldsymbol{R}_b, \boldsymbol{Q}^T, \boldsymbol{I}_n) \in \mathbb{R}^{(n_\xi \times n_\xi)}$, $\boldsymbol{R}_b$ is the rotation matrix from the body frame to the inertial frame, $\boldsymbol{Q} = \boldsymbol{R}_b^T \boldsymbol{R}_b$.

B. Contact Modeling

The contact between the target and the UAM is illustrated in Fig. 2. We assume the contact between the UAM and the target is completely inelastic, and the contact is instant at t_0. Previous research like [15] considered the effect of the caught target as a virtual external force $\boldsymbol{u}_{ext}$ applied on the UAM and estimated the force using the contact model. However, the external force is not suitable for target catching, since the impulse is difficult to estimate. In this letter, we model the contact from the impulse perspective. The target's momentum $\boldsymbol{P}_t = m_t \boldsymbol{v}_{t_{0^-}}$ is the product of the mass of the target m_t and the velocity $\boldsymbol{v}_{t_{0^-}}$ of the target right before impact t_{0^-}. Integrating the bottom n lines of (1) from t_0 to $t_i (t_i > t_0)$, we have the following impact model

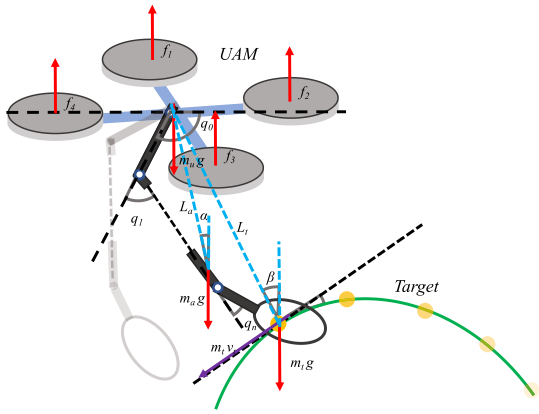


Fig. 2. Target catching model.

$$\begin{aligned} &\int_{t_0}^{t_i} \Big(\boldsymbol{M}_{13}^T \ddot{\boldsymbol{x}}_u + \boldsymbol{M}_{23}^T \ddot{\boldsymbol{\Phi}}_u + \boldsymbol{M}_{33} \ddot{\boldsymbol{q}} + \boldsymbol{C}_1 \dot{\boldsymbol{x}}_u \\ &\quad + \boldsymbol{C}_2 \dot{\boldsymbol{\Phi}}_u + \boldsymbol{C}_3 \dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) \Big) \, \mathrm{d}t \\ &= \int_{t_0}^{t_i} (\boldsymbol{\tau}_a - \boldsymbol{\tau}_{ext}) \, \mathrm{d}t, \end{aligned} \tag{3}$$

where $\boldsymbol{\tau}_a$ is the motor command, $\boldsymbol{\tau}_{ext}$ is the generalize torque introduced by the target. The integration of $\boldsymbol{\tau}_{ext}$ consists of the impulse $\boldsymbol{P}_t$ applied on the UAM and the torque introduced by the target mass, which is

$$\int_{t_0}^{t_i} \boldsymbol{\tau}_{ext} \mathrm{d}t = \boldsymbol{L}_t \boldsymbol{P}_t + \int_{t_0}^{t_i} m_t g \boldsymbol{L}_t \sin\beta \mathrm{d}t, \tag{4}$$

where $\boldsymbol{L}_t$ is the vector from the target to the arm base, and β is the angle between $\boldsymbol{L}_t$ and $\boldsymbol{j}_3 = [0, 0, 1]^T$, as shown in Fig. 2. At t_0, with the instant contact assumption, the target's momentum $\boldsymbol{P}_t$ is transferred to the UAM-target system. After t_0, Equation 4 is simplified as $\int_{t_{0+}}^{t_i} \boldsymbol{\tau}_{ext} \mathrm{d}t = \int_{t_{0+}}^{t_i} m_t g \boldsymbol{L}_t \sin\beta \mathrm{d}t$, which implies $\boldsymbol{\tau}_{ext} = m_t g \boldsymbol{L}_t \sin\beta$ after t_0.

C. Problem Formulation

Denoting the task for the UAM to follow a given trajectory as $\boldsymbol{x}_d, \dot{\boldsymbol{x}}_d, \ddot{\boldsymbol{x}}_d$, the influence of catching a fast-flying heavy target is bilateral. First, the momentum applied to the UAM may destroy the flight stability, and second, the heavy target with an unknown mass influences the controller's performance. The first influence is characterized by the maximum position displacement X_m and maximum angular displacement θ_m, in which $X_m = \max_{t_0 \le t_i, t_j \le t_1} \|\boldsymbol{x}_u(t_i) - \boldsymbol{x}_u(t_j)\|$ and $\theta_m = \max_{t_0 \le t_i, t_j \le t_1} (\|\boldsymbol{\Phi}_u(t_i)\| - \|\boldsymbol{\Phi}_u(t_j)\|)$, t_i, t_j are any two moments in $[t_0, t_1]$. The second influence is characterized by the final tracking error between the given trajectory and the position of UAM, defined as $X_s = \|\boldsymbol{x}_d(t_1) - \boldsymbol{x}_u(t_1)\|$.

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The problem is divided into two decoupled subproblems that need to be addressed:

Problem 1: Prevent the impact applied to the UAM from destroying its flight stability when catching a fast-flying heavy target. This subproblem is equivalent to reducing θ_m and X_m.

Problem 2: Improve the trajectory tracking performance of UAM when catching a heavy target with an unknown mass. This subproblem is equivalent to reducing the tracking error X_s.

The first problem emphasizes the influence of the target's momentum on the UAM, while the second problem emphasizes the influence of the target's mass on the UAM.

III. Method

To solve the problems formulated in the previous section simultaneously, we propose a compensation method combining state estimation and controller design to reduce the influence of external impacts on the flight and ensure safety. First, the target mass is rapidly estimated utilizing the angular momentum theorem. Second, an impedance controller in joint space is designed for the arm to absorb the impact introduced by the target. Third, a compensation term is added to the controller of the UAM to compensate for the target mass and joint movement.

A. Fast Angular-Momentum-Theorem-Based Target Mass Estimation

After the catching stage, target mass m_t can be easily estimated with the total thrust from (1), replacing $(m_u + m_a)$ with $(m_u + m_a + m_t)$. Yet, we aim to estimate the target mass as soon as possible during the catching stage instead of waiting seconds to estimate the mass from the thrust.

Considering (3), since the mass of the UAV base is much higher than the arm, the acceleration of the UAV is way slower than the arm. It is reasonable to neglect the influence of the base movement in the mass estimation process, and (3) is simplified as

$$\int_{t_0}^{t_i} \left(\boldsymbol{M}_{33}\ddot{\boldsymbol{q}} + \boldsymbol{C}_3\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q})\right) \mathrm{d}t = \int_{t_0}^{t_i} \left(\boldsymbol{\tau}_a - \boldsymbol{\tau}_{ext}\right) \mathrm{d}t, \quad (5)$$

Rearranging (5) and (4), and yields

$$\begin{aligned} &\boldsymbol{M}_{33}\dot{\boldsymbol{q}}(t_i) + \boldsymbol{C}_3\boldsymbol{q}(t_i) - \boldsymbol{M}_{33}\dot{\boldsymbol{q}}(t_0) - \boldsymbol{C}_3\boldsymbol{q}(t_0) \\ &= \int_{t_0}^{t_i} \left(\boldsymbol{\tau}_a - m_t g \boldsymbol{L}_t \sin\beta - G(\boldsymbol{q})\right) \mathrm{d}t - m_t \boldsymbol{v}_{t_{0^-}} \boldsymbol{L}_t, \end{aligned} \quad (6)$$

where $\dot{\boldsymbol{q}}(t_{0^+})$ is the angular velocity of joints at t_{0^+}; $\dot{\boldsymbol{q}}(t_i)$ is the angular velocity at t_i.

From (6), the mass of the target at t_i can be expressed as

$$m_t = \frac{\int_{t_0}^{t_i} (\boldsymbol{\tau}_a - G(\boldsymbol{q}))\mathrm{d}t + \boldsymbol{M}_{33}(\dot{\boldsymbol{q}}(t_0) - \dot{\boldsymbol{q}}(t_i)) + \boldsymbol{C}_3(\boldsymbol{q}(t_0) - \boldsymbol{q}(t_i))}{\boldsymbol{v}_{t_{0^-}} \boldsymbol{L}_t + \int_{t_0}^{t_i} g \boldsymbol{L}_t \sin\beta \mathrm{d}t}. \quad (7)$$

In (7), $\boldsymbol{q}, \dot{\boldsymbol{q}}$ are directly measured with motor sensors, $\boldsymbol{L}_t$ are calculated with $\boldsymbol{q}$ and arm link lengthes available offline. Noting that practically, $\boldsymbol{\tau}_a$ is estimated through the relationship between the motor current measurement and torque with $\boldsymbol{\tau}_a = KI$, and $\boldsymbol{v}_{t_{0^-}}$ is estimated by differentiating the target's positions. Since these variables are from different unsynced sensors, measurements with asynchronized timestamps may influence the accuracy; therefore, an incremental estimation of target mass is introduced. By modifying (7), the estimation of target mass is obtained through

$$\widehat{m}_t = \frac{\boldsymbol{\tau}_{int}(k) + \boldsymbol{M}_{33}\left(\dot{\boldsymbol{q}}(0) - \dot{\boldsymbol{q}}(k)\right) + \boldsymbol{C}_3\left(\boldsymbol{q}(0) - \boldsymbol{q}(k)\right)}{\boldsymbol{v}_{t_{0^-}} \boldsymbol{L}_t + GLT(k)} \quad (8)$$

where

$$\begin{cases} \boldsymbol{\tau}_{int}(k) = \boldsymbol{\tau}_{int}(k-1) + (\boldsymbol{\tau}_a - G(\boldsymbol{q}))(t(k) - t(k-1)) \\ GLT(k) = GLT(k-1) + \widehat{m}_t g \boldsymbol{L}_t \sin\beta (t(k) - t(k-1)) \end{cases},$$

in which $\boldsymbol{\tau}_{int}(k)$ is a middle variable indicating the estimation of $\int_{t_0}^{t_i}(\boldsymbol{\tau}_a - G(\boldsymbol{q}))$, $\boldsymbol{\tau}_{int}(0) = 0$, GLT is a middle variable indicating the estimation of $\int_{t_0}^{t} g\boldsymbol{L}_t \sin\beta \mathrm{d}t$, $GLT(0) = 0$, and k is the measurement index. The estimation $\widehat{m}_t$ is updated when a new measurement of $\boldsymbol{q}$ arrives.

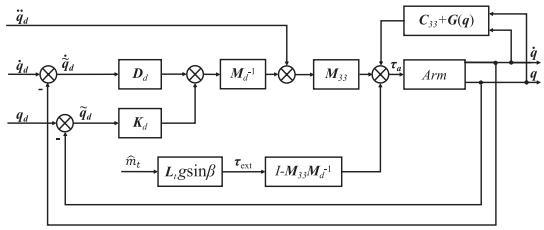


Fig. 3. Impedance controller diagram.

B. Impedance Control for Impact Absorbing

An impedance control method utilizing the estimated target mass is proposed to absorb the impact the target brings; it aims to reduce the stiffness of the manipulator to avoid the bouncing back of the target or causing a significant influence on the attitude and position of the UAV body.

As shown in Fig. 3, the impedance controller is implemented in joint space, whose target is to make the joint a 2nd-order dynamical system to absorb the impact:

$$\boldsymbol{M}_d\ddot{\tilde{\boldsymbol{q}}} + \boldsymbol{D}_d\dot{\tilde{\boldsymbol{q}}} + \boldsymbol{K}_d\tilde{\boldsymbol{q}} = \boldsymbol{\tau}_{ext}, \quad (9)$$

where $\boldsymbol{M}_d, \boldsymbol{D}_d, \boldsymbol{K}_d$ are the parameters of the expected dynamics, $\tilde{\boldsymbol{q}} = \boldsymbol{q}_d - \boldsymbol{q}$ is the error between $\boldsymbol{q}$ and desired joint angle $\boldsymbol{q}_d$.

The influence of momentum in Equation 4 is considered in the contact model. As described in Section II-B, we have $\boldsymbol{\tau}_{ext} = m_t g \boldsymbol{L}_t \sin\beta$ after the contact instance t_0. To make the joints dynamics as (9), the motor command $\boldsymbol{\tau}_a$ is designed as

$$\begin{aligned} \boldsymbol{\tau}_a = &\boldsymbol{M}_{33}\ddot{\boldsymbol{q}}_d + \boldsymbol{C}_3\dot{\boldsymbol{q}} + G(\boldsymbol{q}) + \boldsymbol{M}_{33}\boldsymbol{M}_d^{-1}\left(\boldsymbol{D}_d\dot{\tilde{\boldsymbol{q}}} + \boldsymbol{K}_d\tilde{\boldsymbol{q}}\right) \\ &+ \left(\boldsymbol{I}_n - \boldsymbol{M}_{33}\boldsymbol{M}_d^{-1}\right)\widehat{m}_t g \boldsymbol{L}_t \sin\beta. \end{aligned} \quad (10)$$

Since the UAV is much heavier than the arm, the acceleration of UAV motion is much slower than the arm, and we neglect the influence of the UAV movement on the impedance controller. The UAV's base movement during the impedance controller absorbing the impact is small and can be neglected. Taking (10) and $\boldsymbol{\tau}_{ext} = m_t g \boldsymbol{L}_t \sin\beta$ into (5), we have

$$\boldsymbol{M}_{33}\ddot{\boldsymbol{q}} + \cancel{\boldsymbol{C}_3\dot{\boldsymbol{q}}} + \cancel{G(\boldsymbol{q})} = \boldsymbol{M}_{33}\ddot{\boldsymbol{q}}_d + \cancel{\boldsymbol{C}_3\dot{\boldsymbol{q}}} + \cancel{G(\boldsymbol{q})}$$

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$$+ \boldsymbol{M}_{33}\boldsymbol{M}_d^{-1}\left(\boldsymbol{D}_d\dot{\tilde{\boldsymbol{q}}} + \boldsymbol{K}_d\tilde{\boldsymbol{q}}\right)$$
$$+ \left(\boldsymbol{I}_n - \boldsymbol{M}_{33}\boldsymbol{M}_d^{-1}\right)\widehat{m}_t g\boldsymbol{L}_t \sin\beta - m_t g\boldsymbol{L}_t \sin\beta. \quad (11)$$

Assuming that the target estimation is accurate, we have

$$\cancel{\boldsymbol{M}_{33}}\boldsymbol{M}_d^{-1} m_t g\boldsymbol{L}_t \sin\beta = \cancel{\boldsymbol{M}_{33}}\ddot{\tilde{\boldsymbol{q}}} + \cancel{\boldsymbol{M}_{33}}\boldsymbol{M}_d^{-1}\left(\boldsymbol{D}_d\dot{\tilde{\boldsymbol{q}}} + \boldsymbol{K}_d\tilde{\boldsymbol{q}}\right). \quad (12)$$

Multipling both sides of (12) with $\boldsymbol{M}_d$ yields (9), which is the desired closed-loop dynamics of the arm.

C. Mass Compensation for Target Influenced UAM Control

The target introduces influence in lift and torque, which need to be compensated separately. This letter modified the widely-used geometric controller [33] by incorporating the estimated target mass $\widehat{m}_t$, compensating for both the influence of target gravity and the effect on the torque. The UAM controller outputs are as follows:

$$\begin{aligned} f_u\boldsymbol{b}_3 = & -\left(-k_x\boldsymbol{e}_x - k_v\boldsymbol{e}_v - (m_u + m_a + \widehat{m}_t)g\boldsymbol{j}_3\right. \\ & \left.+(m_u + m_a + \widehat{m}_t)\ddot{\boldsymbol{x}}_d\right)\cdot R\boldsymbol{j}_3, \\ \boldsymbol{\tau}_u = & -k_R\boldsymbol{e}_R - k_\Omega\boldsymbol{e}_{\boldsymbol{\Omega}} + \boldsymbol{\Omega}\times J'\boldsymbol{\Omega} \\ & - J'\left(\hat{\boldsymbol{\Omega}}R^TR_d\boldsymbol{\Omega}_d - R^TR_d\dot{\boldsymbol{\Omega}}_d\right) + \widehat{m}_t g\boldsymbol{L}_t\sin\beta, \end{aligned} \quad (13)$$

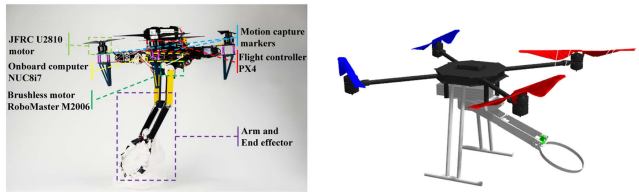
where $\boldsymbol{b}_3$ is the direction of the thrust force, $\boldsymbol{e}_x = \boldsymbol{x}_d - \boldsymbol{x}_u, \boldsymbol{e}_v = \dot{\boldsymbol{x}}_d - \dot{\boldsymbol{x}}_u, \boldsymbol{e}_R$ is the rotation error, $\boldsymbol{e}_\Omega$ is the angular velocity error, k_x, k_v, k_R, k_Ω are constants, $J' = J + J_t$ is the combination of the UAM momentum J and the target momentum $J_t = \widehat{m}_t\boldsymbol{L}_t^2$.

IV. Validation

This section presents simulations and experiments to validate the effectiveness of the proposed approach. Evaluation metrics θ_m, X_m, X_s are analyzed to illustrate the reliability and feasibility of the proposed approach, denoted as AIC_MC, which consists of the mass estimation, mass compensated Arm Impedance Controller, and Mass Compensation algorithm for UAM attitude controller. For the ablation experiment, we define the approach with mass estimation and mass compensated Arm Impedance Controller as AIC. To evaluate the performance of the proposed approach, we compare it with two baseline approaches. Baseline1 is defined as the combination of the geometric controller [33] without mass compensation for the UAV base and position controller for arm joints. The whole-body impedance controller [32] is also implemented as Baseline2 for comparison.

A. Validation Platform

An aerial manipulator platform is designed to validate the proposed algorithms. The platform comprises UAM prototypes realized in both the simulation and the real world, as shown in Fig. 4(a) and (b). The weight of the UAM is 2.4 kg, including the battery and the arm($m_a = 0.2$kg). The arm is driven by brushless motors (RoboMaster M2006), which take current as input and provide angle, angular velocities, and current feedback. Its gear ratio is 54:1, much lower than the typical servo, making implementing the impedance controller much easier [4]. A net is mounted to the arm as the end effector to simplify the catching process. The real-world experimental setup consisted of an OptiTrack motion capture system for position and orientation feedback at 120 Hz and a WiFi router for communication. The simulation setup combines Pixhawk + Gazebo as software in the loop (SITL), which accurately and efficiently simulates UAM's behavior. Since there are no motor current measurements in the Gazebo, we mounted a force/torque sensor on the joint instead. The main parameters of the UAM, including UAM mass and arm length, are consistent in the real UAM platform and the simulation environment.



(a) Aerial manipulator. (b) Simulated aerial manipulator.

Fig. 4. Experimental and simulational platform.

TABLE I
ESTIMATION FOR DIFFERENT TARGET MASS

Mass(kg)	Error(%)	STD
0.01	46.16	0.002
0.05	4.46	0.001
0.1	7.01	0.007
0.3	7.95	0.028

TABLE II
ESTIMATION FOR DIFFERENT VELOCITY

Vel(m/s)	Error(%)	STD
1.0	9.93	0.004
2.0	8.13	0.006
5.0	4.46	0.002
10.0	3.65	0.003

B. Simulations

1) Mass Estimation: The target mass is estimated according to (8). The target mass is estimated within three or four measurements of q. Because the joint angle vector q is sampled at 50 Hz, and thus the estimation time is about 0.08 s. Two simulations are designed to validate the mass estimation method with different target masses and velocities.

In the first simulation, the target velocity was set to 5 m/s, and the target mass was set to 0.01-0.3 kg. Each target was thrown ten times. The mass estimation results are shown in Table I. The estimation errors in the 0.05–0.3 kg range are smaller than 10%, and the estimation error of the 0.01 kg ball is 46.16%, which might be because the influence of additive uncertainty on a smaller mass target is more significant than a higher mass target.

In the second simulation, the target mass was set to 0.05 kg, and the target was thrown to the UAM at different velocities(1.0–10.0 m/s) ten times. The estimation error in Table II shows that

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TABLE III
POSITION AND ORIENTATION DISPLACEMENTS UNDER DIFFERENT METHODS

test	Baseline1			Baseline2			AIC			AIC_MC		
	θ_m(rad)	X_m(m)	X_s(m)	θ_m(rad)	X_m(m)	X_s(m)	θ_m(rad)	X_m(m)	X_s(m)	θ_m(rad)	X_m(m)	X_s(m)
1	0.122	0.183	0.090	0.113	0.185	0.105	0.061	0.123	0.114	0.059	0.105	0.055
2	0.168	0.171	0.102	0.154	0.198	0.139	0.070	0.106	0.101	0.062	0.120	0.061
3	0.152	0.165	0.125	0.078	0.219	0.117	0.070	0.152	0.123	0.083	0.051	0.028
4	0.132	0.124	0.108	0.119	0.195	0.143	0.074	0.141	0.080	0.057	0.086	0.038
5	0.076	0.123	0.121	0.103	0.223	0.099	0.063	0.157	0.133	0.084	0.057	0.046
avg	0.130	0.153	0.109	0.115	0.204	0.121	**0.068**	0.136	0.110	0.069	**0.084**	**0.046**

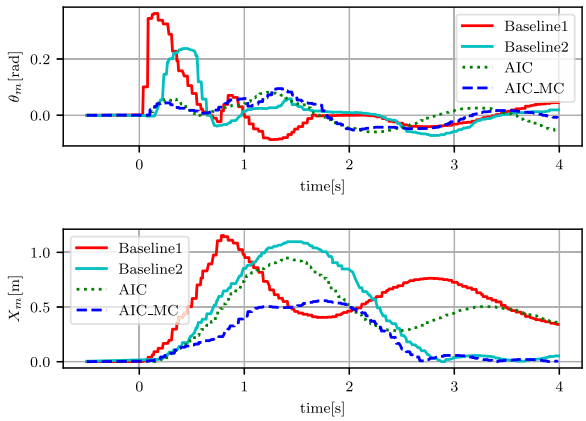


Fig. 5. Position and orientation displacements of UAM under the baselines, AIC, and AIC_MC methods in simulation, where $t_0 = 0$.

the mass estimation error is lower than 10% in an extensive velocity range.

The simulations above show that the mass estimation method is effective and robust to different target masses and velocities. The estimation error is lower than 10% in most cases, which is acceptable for the impedance controller and mass compensation algorithm.

2) Impedance Controller and Mass Compensation: The arm joints' torque feedback in the simulation is from the force/torque sensor mounted on each arm joint. Without loss of generality, a 0.3 kg target was required to be caught by the UAM at 5 m/s five times. The evaluation metrics θ_m, X_m, X_s of baselines, AIC, and AIC_MC are shown in Table III. Compared with the performance of Baseline1, the AIC reduced the θ_m for **48.01%**, X_m for **11.34%**. AIC failed to reduce the final tracking error X_s(0.109 m for Baseline1 and 0.110 m for AIC), which is because the X_s is mainly introduced by the UAM position controller. The improvement on θ_m of AIC_MC is nearly identical to AIC and is **47.03%** lower than Baseline1. The X_m of AIC_MC is **45.43%** lower than Baseline1. The most significant improvement of AIC_MC against AIC and Baseline1 is the X_s, which reduces the average tracking error from 0.109 m to 0.041m(**62.69%**). In general, AIC_MC shows the best performance in maintaining the stability of the UAM.

The position and orientation displacements of UAM under the baselines, AIC, and AIC_MC approaches catching targets of 0.3 kg at 10 m/s are shown in Fig. 5. Compared with baselines, the proposed AIC_MC significantly reduced the position and orientation displacements.

TABLE IV
ESTIMATION FOR DIFFERENT TARGET

Mass(kg)	Estimation(kg)	Error(%)	STD
0.0975	0.0980	5.95	0.0081
0.234	0.2158	8.65	0.0094

To push the limit of the proposed algorithm, the UAM was required to catch a target weighing 0.6 kg (25% mass of the UAM). The supplementary video shows that when a 0.6 kg target was thrown to the UAM at 10 m/s, the UAM under the baselines lost balance and crashed, and the UAM under the proposed AIC_MC remained stable. The simulation illustrated that the proposed approach could maintain the stability of the UAM under the impact of a fast-flying heavy target.

C. Real-World Experiments

Two experiments on the real-world platform were designed to validate the proposed approach. The first experiment required the UAM to catch different targets to validate the mass estimation algorithm. In the second experiment, to better evaluate the performance of the proposed approach, targets of different masses were thrown to UAM under the baselines, AIC, and AIC_MC methods.

1) Mass Estimation: We tested readily available targets at 4 m/s, and a target is shown in Fig. 6. The mass of targets is 0.0975 kg and 0.234 kg, respectively, and each target is thrown ten times. The estimation result is shown in Table IV. Estimation errors of each target are **5.95%** and **8.65%**. The estimation error is slightly larger than in the simulation because the output torque feedback is calculated from the inaccurate current measurements.

2) Impedance Controller and Mass Compensation: The snapshots of an experiment are shown in Fig. 6. The first and last subfigures demonstrate that the UAM returned to the vicinity of its initial position. Fig. 7 shows the position and angular displacement of UAMs under the baselines, AIC, and AIC_MC methods when catching a 0.234 kg target. The evaluation metrics θ_m, X_m, X_s under the different methods are shown in Table V. The maximum angular displacement and position displacement of the baselines are similar. However, Baseline2 produced the lowest position error X_s, which shows the effectiveness of the whole body impedance controller. It is clear that the AIC effectively reduces the maximum angular displacement θ_m by **45.63%**, compared with the UAM under the Baseline1. The

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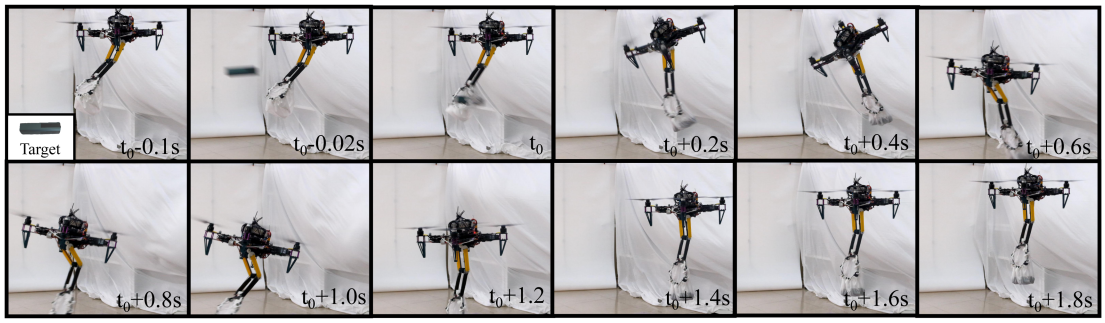


Fig. 6. Snapshots of UAM catching a target and absorbing the impact under the proposed AIC_MC.

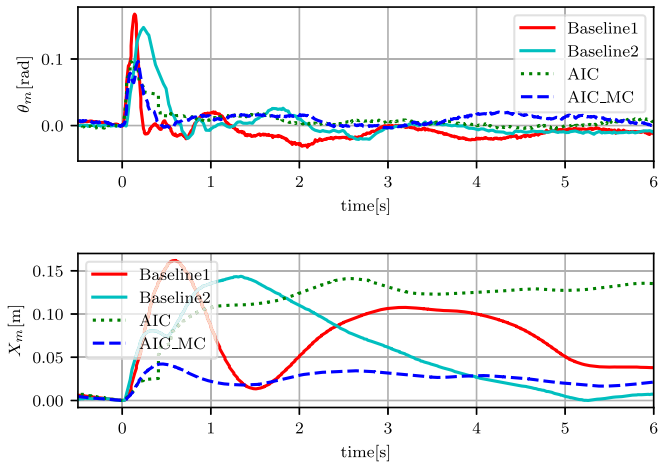


Fig. 7. Position and orientation displacements of UAM under the baselines, AIC, and AIC_MC methods in the real-world experiment, where $t_0 = 0$.

TABLE V
POSITION AND ANGULAR DISPLACEMENTS OF UAM UNDER DIFFERENT METHODS, WITH THE MINIMUM VALUES BOLDED

	θ_m(rad)	X_m(m)	X_s(m)
Baseline1	0.1822	0.1751	0.1382
Baseline2	0.1440	0.1429	**0.0067**
AIC	0.0991	0.1504	0.1396
AIC_MC	**0.0904**	**0.0470**	0.0256

AIC_MC further reduces the maximum position displacement X_m by **73.18%**, compared with the UAM under the Baseline1. The AIC_MC reduced position error X_s by **81.51%** compared with Baseline1. In reducing final tracking error X_s, the performance of the proposed algorithm is not as good as Baseline2. And the X_s of AIC_MC (0.0256 m) is close to the performance of the Baseline2(0.0067 m), compared with the Baseline1(0.1382 m). AIC_MC converged to the desired pose faster than Baseline1 and Baseline2.

The UAM catches targets of different weights to demonstrate the effectiveness and robustness of the proposed approach. Fig. 8 shows the position and angular displacement of UAM under the baselines, AIC, and AIC_MC catching targets weighing 0.0975 kg, 0.160 kg, and 0.234 kg, repeated five times. The effectiveness of the proposed approach is illustrated through

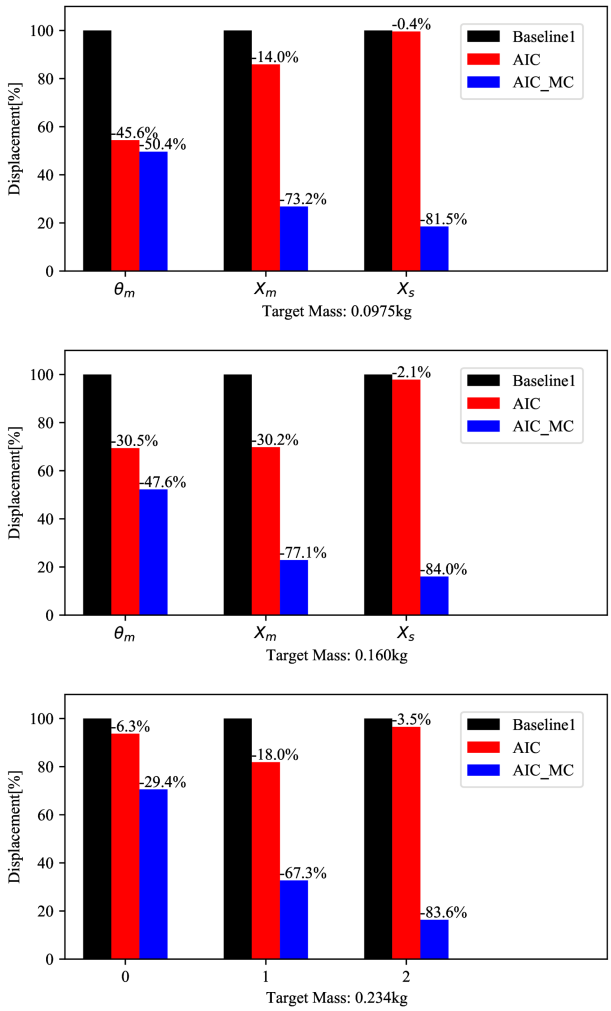


Fig. 8. Average position and orientation displacements of UAM in the experiment while catching targets of different weights.

the experiments. The AIC can effectively reduce the angular displacement of the UAM in the catching stage, and the AIC_MC can suppress both position displacement and the tracking error of the UAM.

As demonstrated in the supplementary video, we conducted an experiment where a Coke weighing 0.356 kg (equivalent to **14.8%** of the UAM's mass) was thrown toward the UAM.

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This experiment aimed to showcase the UAM's capability to effectively absorb the impact generated by a significantly heavy target. Since 75% of the total lift of the UAM was used to overcome its gravity, the heavy target is lighter than that (**25%** of the UAM's mass) in the simulation. Remarkably, the UAM maintained stability after catching the Coke can.

V. Conclusion

This letter develops a novel approach to address the problem of aerial catching stability for fast-flying heavy targets (up to 25% of the UAM mass at 10 m/s). We propose a fast target mass estimation algorithm to estimate the target mass within 0.1 seconds. We design a novel impedance controller for the UAM arm to effectively absorb the target's kinetic energy. We develop a compensation algorithm for the UAM position controller based on the estimated target mass. Combining the above components, we propose the **AIC_MC** approach for aerial catching stability. Simulations and experiments show that the proposed approach can effectively absorb the impact of the target and reduce the angular/position displacement of the UAM. The proposed approach can be applied to catch a heavy target in the real world while maintaining the flight safety of the UAM.

Several directions are ongoing research. The first is to use the proposed approach to target catching while UAM moves. The second is to generalize the algorithm to arm with more degrees of freedom, enabling more complex missions such as faster and heavier target catching, batting, juggling, etc. The third is to reduce the platform weight for mobility enhancement and finally reach the target/UAM mass ratio in the natural environment, which can be 20–30%.

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